

Qingyang Zhang

stevenz3@andrew.cmu.edu • +1 412-657-6580 • www.linkedin.com/in/qingyazhang

EDUCATION

Carnegie Mellon University

M.S. in Information Networking

Pittsburgh, United States

August 2025 - May 2027

New York University

B.S. in Computer Science & Data Science

GPA: 3.9/4.0

New York, United States

August 2021 - May 2025

EXPERIENCE

Carnegie Mellon University

May 2025 - Aug 2026

Machine Learning Researcher — Cross-National EHR Representation Learning & Domain Adaptation

- Built scalable PySpark/Slurm pipelines for longitudinal healthcare data processing and leakage-controlled prediction tasks across 3.98M patients, enabling distributed ML experimentation and evaluation.
- Developed a BoW similarity + class-ratio data selection strategy for domain-adaptive foundation-model fine-tuning, preserving all positives while selecting the most relevant external negatives.
- Engineered a clinical ontology alignment pipeline across 7 medical coding systems using deterministic mappings and embedding-based FAISS retrieval.
- Implemented Asymmetric Supervised Contrastive Learning for temporal EHR representations and evaluated NHIRD→MIMIC-IV/EHRSHOT transfer, achieving the best AUPRC on 3/4 controlled ablations and outperforming published baselines at 16-shot across five incident-disease tasks.

NYU Grossman School of Medicine

May 2024 - Aug 2025

Machine Learning Researcher — Automated Radiology Report Generation

- Built a multimodal radiology report generation pipeline combining DenseNet/Swin Transformer vision encoders with GPT-4 and Llama 3.
- Fine-tuned medical vision-language models (MAIRA-1/2) on knee X-rays for automated radiology report generation.
- Collaborated with physicians to design structured report templates and improve the clinical relevance of model-generated reports.
- Built image preprocessing pipelines using OCR, erosion, and dilation to remove artifacts and reduce noise in radiology images.

NYU Grossman School of Medicine

May 2024 - Aug 2024

Machine Learning Researcher — Clinical NLP / Pathology Extraction

- Built an NLP pipeline for extracting pathologies from radiology reports using pretrained and fine-tuned BERT-based medical language models (NYUTron).
- Fine-tuned Llama 2/3 with LoRA for parameter-efficient adaptation to clinical text.
- Improved exposure to key pathologies for medical residents by 135% through model-assisted case identification.

New York University

Sep 2023 - Aug 2024

Research Assistant — Term Extraction & NLP

- Led the development of the Chinese version of the Termolator project, which entailed employing keywords to systematically search for relevant terms on Wikipedia.
- Designed an evaluation framework for terminology extraction without explicit ground-truth labels by comparing results across independent encyclopedia sources.
- Applied KNN and graph-based label propagation techniques to refine extracted terminology and improve glossary quality.

TECHNICAL SKILLS

Languages: Python, C++, Java, R, SQL

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, Vision-Language Models, Contrastive Learning, LoRA, Embeddings, FAISS

Data & Systems: PySpark, Pandas, NumPy, Docker, Slurm, Git, Linux, MySQL

ML Domains: Multimodal Learning, Computer Vision, NLP, LLMs, Healthcare ML, Representation Learning, Retrieval